

The research team used Bose-Einstein condensate to make atomtronic sensors. The team reports creating this gas by cooling sodium atoms suspended in magnetic fields. Researchers then trapped the atoms in a pair of crossed laser beams and further chilled the atoms to less than 10-billionths of a degree above absolute zero. The two beams also shaped the condensate into a flattened dough-nut with a radius of about 20 micrometre.

A second pair of lasers transferred energy to the dough-nut to start its rotation. Because atoms in the condensate behave as a single, coherent quantum particle, such a ring of the substance does not speed up or slow down gradually. It jumps between different speeds, much like a blender would, if it could change settings instantaneously.

The scientists chose the lowest setting for their ring; about one revolution every second. Because the condensate also happens to be frictionless, this ring should, in theory, rotate forever. Limited by technical difficulties, the research team kept it going for about 40 seconds—the lifetime of their condensate.

Scientists believed that Bose-Einstein condensate could provide an extremely sensitive rotation sensor. They added a 'weak link' to their condensate ring—a barrier created by a blue laser that could speed up or shut down the flow. Theoretically, if the condensate were kept still and the barrier was attached to a rotating sensor, the barrier would cause a sudden jump in current at certain rotation speeds.

Atomtronic devices

The atoms in the condensate flow as a current, which can be switched on and off like a normal circuit. Atomtronics uses atoms in strange quantum states to power devices or computer memory. This is different from spintronics, which stores information based on the spin of individual electrons, allowing each one to store two bits of data instead of one.

Computer scientists and particle physicists have made several ad-

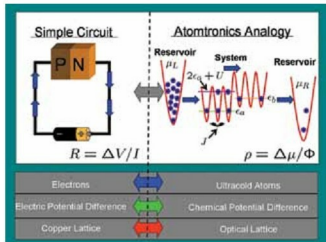


Fig. 2: Atomtronic analogy to diode circuit

vances in these fields in the past several months. Using atoms instead of electrons to process information could change the way we think about computing. In quantum computing we store a quantum state on an object, perform operations on the object and then read out the final state. If the system is not coherent, the initial stored information is lost.

Atoms trapped in optical lattices have been considered extensively for specific quantum computing schemes due to their inherent energy conserving characteristics. Therefore the dynamics of atomtronic devices would be coherent and potentially useful in quantum computing. It is also suggested that there is the possibility that atomtronics could be useful in obtaining sensitive measurements. It is thus concluded that atomtronic systems provide a nice test of fundamental concepts in condensed matter physics. While these ideas have been modeled, they are yet to be built.

Atomtronic diode

Atomtronic diode is a device that allows an atomic flux to flow across it in essentially only one direction. The atomtronic analogy of a diode is formed from the joining of p- and n-type semiconductor materials. Electrons are replaced by ultra-cold atoms, the battery is replaced by high and low chemical potential reservoirs,

and the metallic crystal lattices (the microscopic medium that the electrons traverse) are replaced by an optical lattice. The atomtronic diode is achieved by energetically shifting one-half of the optical lattice with respect to the other.

The wires and atomtronic components are composed

of optical lattices, and current refers to the number of atoms that pass a specific point in a given amount of time. The desired function of an atomtronic transistor is to enable a weak atomtronic current to be amplified, or to switch on or off a much larger one. The team has also modeled an atomtronic transistor. The atomtronic version of transistor exhibits on/off switching behavior and acts as an amplifier.

By configuring the optical lattice in a manner researchers show that it is possible to recover the characteristics of the conventional electronic transistor in the atomic world.

Limitations of atomtronics

Scientists are hoping to use the condensate in the way that superconductors have been used to make improved devices and sensors. Idea for a useful device was inspired by superconducting quantum interference devices, commonly known as SQUIDS. Scientists also believe that Bose-Einstein condensate could provide an extremely sensitive rotation sensor.

It is pointed out, however, that atomtronics probably won't replace electronics as atoms are sluggish compared to electrons. This means it might be difficult to replace fast electronic devices with sluggish atomtronic devices. ●

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